

Test Plan — SCRUM-10: Requirement for cart module

Field	Value
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Ticket	SCRUM-10 — https://tnplbharat.atlassian.net/browse/SCRUM-10
Title	Requirement for cart module
Type / Status	Story / In Review
Priority	Medium
Project	QA Testing
Sprint / Fix version	Not set on the ticket
Components / Labels	None
Linked issues / Sub-tasks / Attachments	None
Created / Updated	2026-09-12
Source	Jira Cloud REST API v3 (/rest/api/3/issue/SCRUM-10), fetched 2026-09-12
Document status	DRAFT — awaiting human sign-off

This plan is a draft, not an approved artifact. It is derived from the ticket description alone. The ticket has **no acceptance-criteria section**, so scenarios are traced to description statements and to the gaps in Section 2 — not to agreed ACs. Nothing here is final until a human approves it.

1. Scope & Objectives

In scope (as listed on the ticket):

- Cart operations: add item, add multiple quantity, add multiple items, remove item, update quantity, empty cart.
- Cart persistence: across page refresh, and (conditionally) across tabs/windows.
- Pricing: subtotal, tax, shipping cost, total.
- Cart UI: line display, totals display, responsive layout, call-to-action buttons, UI error messages.
- Advanced (conditionally in scope): coupons, save-for-later, guest checkout, payment-gateway hand-off.
- Edge and negative cases: out of stock, large quantities, zero quantity, invalid input, session timeout.

Objective: confirm the cart computes and displays monetary values correctly for every mutation and boundary condition, and that it degrades safely on invalid input, out-of-stock items, and session loss.

Highest-risk area is money. Subtotal / tax / shipping / total arithmetic, rounding precision, and whether server-side totals trust client-supplied quantities and prices. A defect here is financial and reputational, so price-integrity cases are rated **P0** below even though the ticket files calculations under "High Priority".

Major unknown: the application under test, its environment, and its test data are not defined anywhere (G3–G6). This repository contains no cart implementation, so the module under test is external to it.

2. Gaps & Questions for the author

Scored  present ·  ambiguous ·  missing. Every / below is a question the author needs to answer before execution. **This section is the most valuable output of this plan** — it is cheaper to answer these now than to find the defects in production.

ID	Item	Score	Question for the author
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G 1	Acceptance criteria		The story has no AC section. The description is a list of "verify that..." statements with no expected results, so nothing is objectively pass/fail. Will you supply ACs, or approve the traceability in Section 3 as a substitute?
G 2	User story / goal		There is no "As a ... I want ... so that ...". What is the user goal and the business outcome the cart must deliver?
G 3	Application & environment		Which product, URL, build and browser matrix is under test, and how is the build deployed for QA?
G 4	Test data		No catalog, prices, currency, tax rates, shipping rules or coupon codes are specified. Who supplies them, or should QA seed them?
G 5	Preconditions / setup		Is a pre-seeded cart, a registered account, or a specific session configuration required to begin?
G 6	External integrations		The shipping API and payment gateway are referenced but not named. Which sandbox endpoints and test modes apply?
G 7	"Remove the last item clears the cart (if applicable)"		Is the empty-cart state a requirement, and what should it look like?
G 8	Cross-tab persistence "(if applicable – depends on your implementation)"		Is cross-tab consistency in scope for this release, and should it be immediate or eventual?
G 9	Multi-currency "if your system supports multiple currencies"		Single- or multi-currency? Which currencies, and what rounding/exchange rule?
G 10	Guest checkout "If guest checkout is supported"		Is guest checkout in scope? Does a guest cart merge into the account on login?
G 11	Advanced features "Low Priority ... depending on requirements"		Are coupons, save-for-later, guest checkout and gateway integration in scope for this release, or explicitly deferred?
G 12	Unobservable wording		"doesn't break the cart functionality", "accurately", "handle gracefully" — what is the observable expected result in each case?
G 13	Quantity bounds		What are min/max quantity, per-item stock cap and total-cart cap? Is a quantity of 0 an error or a removal?
G 14	Non-functional requirements		No performance, security/authorization, accessibility, i18n or logging requirements. Which apply? In particular: can a user read or modify another user's cart (cart-id authorization)?
G 15	Regression surface		Which existing pages and flows are affected (product page, mini-cart, checkout, order history, promotions)?
G 16	Designs / mockups		No mockups are linked for the cart, the empty state, or the error states.

ID	Item	Score	Question for the author
G 17	Ownership & stability	⚠	Reporter, assignee, components, sprint and fix version are all empty, and the status is "In Review". Who answers these questions, and is scope frozen?
G 18	Money rounding & precision	✗	Rounding rule per line vs per cart, tax rounding, and currency decimal handling are unspecified.
G 19	Concurrency & staleness	✗	Behaviour when the same cart is mutated from two sessions, when stock depletes mid-session, or when a price changes while an item sits in the cart.

Checklist roll-up

Checklist area	✅	⚠	✗
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Functional	0	4	2
Data & environment	0	1	3
Non-functional	0	1	4
Cross-cutting	0	1	3
Clarity	0	2	1
Total	0	9	13

Nothing on the checklist scored ✅. The ticket describes what to poke at, not what "correct" means.

3. Test Scenarios

Priorities are a **proposal**. The ticket labels its own groups Must-Have / High / Medium / Low / Important; the mapping keeps that intent but promotes money-integrity cases to P0.

- **P0** — blocks release: core cart operation broken, or financial/data integrity at risk.
- **P1** — significant functional or UX impact.
- **P2** — low impact, or scope-dependent (see G11).

Trace references description sections (D-I...D-V) or gaps (G#). Where the expected result depends on an unanswered question it is marked **Spec needed**, not invented.

3.1 Core cart operations — P0 (D-I)

ID	Scenario	Expected result	Pr i	Trace
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C-01	Add one item to an empty cart	Cart holds 1 line, qty 1, line subtotal = unit price	P0	D-I
C-02	Add the same item again	Existing line qty increments to 2; no duplicate line	P0	D-I
C-03	Add the same item with an explicit quantity of 5	qty 5, subtotal = unit price × 5	P0	D-I

ID	Scenario	Expected result	Pr i	Trace
C-04	Add two different items	Two distinct lines with correct independent quantities	P 0	D-I
C-05	Add to cart from the product page	Cart reflects the item and quantity; cart count/badge reflects it	P 0	D-I
C-06	Remove one line from a multi-line cart	That line is gone, other lines unchanged, totals recomputed	P 0	D-I
C-07	Decrease quantity on a line with qty > 1	Quantity drops by 1; totals recomputed	P 0	D-I
C-08	Remove the only/last item	Cart reaches empty state per G7 — Spec needed	P 0	D-I, G7
C-09	Click "empty cart" on a populated cart	All lines removed, totals zero	P 0	D-I
C-10	Increase quantity on an existing line	Line subtotal and cart totals increase correctly	P 0	D-I
C-11	Refresh the page with items in cart	Items and quantities retained; totals recompute correctly	P 0	D-I
C-12	Open the cart in a second tab after adding in the first	Consistency behaviour per G8 — Spec needed	P1	D-I, G8
C-13	Mutate the cart, then log out and back in	Cart is account-bound or session-bound per G14 — Spec needed	P1	D-I, G14

3.2 Price & calculation integrity — P0 (D-II)

ID	Scenario	Expected result	Pr i	Trace
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P-01	Subtotal of a multi-line cart	Equals $\Sigma(\text{unit price} \times \text{qty})$ for all lines	P 0	D-II
P-02	Totals after each mutation (add / remove / qty change / empty)	Always recomputed; never stale	P 0	D-II
P-03	Quantity reduced to 1 (boundary)	Line subtotal equals a single unit price	P 0	D-II
P-04	Tax on a taxable cart	Matches a hand-calculated expected value for the configured rate	P 0	D-II, G4
P-05	Shipping cost per method	Matches the rule for weight / destination / method	P 0	D-II, G6
P-06	Total price	Equals subtotal + tax + shipping – discounts, verified by hand	P 0	D-II
P-07	Rounding & precision	Consistent per-line/per-cart rounding; 2 dp currency; no float artefacts (e.g. 0.1 + 0.2)	P 0	G18
P-08	Zero-value cart (100% coupon, or a zero-price item)	Total is 0, never negative	P1	D-II, D-IV

ID	Scenario	Expected result	Pr i	Trace
P-09	Mixed-currency or currency switch	Consistent totals per G9 — Spec needed	P2	D-II, G9

3.3 Cart UI — P1 (D-III)

ID	Scenario	Expected result	Pr i	Trace
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U-01	Cart line rendering	Each line shows name, unit price, quantity, line subtotal	P1	D-III
U-02	Totals display	Totals shown clearly and match the computed values	P1	D-III
U-03	Responsive layout at 375 / 768 / 1440 px	Usable at all widths; no overflow or clipped controls	P1	D-III
U-04	CTA buttons: Add to Cart, Update Quantity, Remove Item, Checkout	Each performs its stated action	P1	D-III
U-05	Update Quantity with a typed value	Applied on blur/enter; invalid value rejected per G13	P1	D-III, G13
U-06	Add an out-of-stock item	User-visible error message; cart unchanged	P1	D-III
U-07	Empty cart representation	Empty-state message shown; Checkout unavailable	P1	D-III, G7
U-08	Rapid repeated CTA clicks	No double submission or duplicated lines; controls disabled while in flight	P1	D-III, D-V
U-09	Keyboard-only and screen-reader use of cart controls	All controls reachable, labelled and operable	P2	G14
U-10	Browser matrix (Chrome / Edge / Firefox / Safari)	Behaviour consistent per the matrix in G3	P1	D-III, G3

3.4 Edge cases & negative testing — P1 (D-V)

ID	Scenario	Expected result	Pr i	Trace
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E-01	Item in cart goes out of stock before checkout	Clear messaging; checkout blocked for that line	P0	D-V
E-02	Quantity set to 0	Error or removal, per G13 — Spec needed	P1	D-V, G13
E-03	Negative quantity / decrement below 1	Blocked or clamped; never a negative total	P0	D-V
E-04	Non-numeric quantity input (abc , 1e3 , 1.5 , emoji, whitespace, 1000-char string)	Rejected with a message; cart unchanged; no server error	P1	D-V

ID	Scenario	Expected result	Pr i	Trace
E-05	Very large quantity (999 / 10,000 / 999,999)	No overflow, NaN or Infinity; accepted or capped per G13	P1	D-V, G13
E-06	Quantity above available stock	Clamped or blocked with a message	P1	D-V, G13
E-07	Session times out, then the user returns	Cart intact (per D-V) or defined behaviour per G14	P1	D-V
E-08	Cookies/session storage cleared manually	Defined outcome: empty cart, or server-side cart restored	P2	G14
E-09	Concurrent add/remove from two sessions	No duplicate lines, no lost updates	P1	G19
E-10	Product price changes while an item sits in the cart	Cart reprices or holds the original price per G19 — Spec needed	P2	G19
E-11	Tampered request (negative qty, client-supplied price or total)	Server rejects and recomputes authoritatively	P0	G14
E-12	Injection attempts via quantity or coupon fields (SQLi / XSS)	Input sanitised and escaped; no script execution	P0	G14
E-13	Locale or currency changed mid-session	Totals and formatting remain consistent	P2	G9

3.5 Advanced features — P2 (D-IV, pending G11)

ID	Scenario	Expected result	Pr i	Trace
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A-01	Apply a valid coupon	Discount applied to the correct base (subtotal vs total)	P2	D-IV
A-02	Apply an invalid, expired or already-used coupon	Clear error; no discount applied	P2	D-IV
A-03	Coupon boundaries: minimum spend, 0%, 100%, stacked	Behaviour matches the discount rules in G4	P2	D-IV, G4
A-04	Save cart for later, then return	Cart restored with the same items and quantities	P2	D-IV
A-05	Guest checkout end to end	Order completes without account creation (scope per G10)	P2	D-IV, G10
A-06	Guest cart merged into an account at login	Merge behaviour defined, no duplicates or lost items	P2	G10
A-07	Cart handed to the payment gateway	Items, quantities, amounts and currency transmitted correctly	P2	D-IV, G6
A-08	Successful payment	Cart cleared, order created with correct status	P2	D-IV
A-09	Failed or cancelled payment	Cart preserved; no order and no partial order written	P2	D-IV

ID	Scenario	Expected result	Pri	Trace
A-10	Double-click / refresh at the payment step	No duplicate order (idempotency)	P2	D-IV

3.6 Cross-role & permission paths — blocked by G14

ID	Scenario	Expected result	Pri	Trace
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R-01	User A requests User B's cart id directly	Denied; no data leaked	P0	G14
R-02	Guest session accesses a registered user's cart	Denied	P0	G14
R-03	Each role (guest / registered / admin) views cart actions	Only permitted actions are offered and accepted	P1	G14

These cannot be executed until G14 is answered — the authorization model is undefined.

4. Test Data & Environment

4.1 Environment (blocked — G3)

Not specified on the ticket. Needed before execution:

- Application URL and build/version under test; how the build is deployed.
- Browser and device matrix.
- Whether the shipping API and payment gateway have usable sandbox modes (G6).
- Access to logs / network inspection for verifying server-side totals.

4.2 Test data set required (blocked — G4)

To be supplied or seeded. Every row below is a request to the author, not a confirmed fixture.

Entity	Variants needed	Status
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Products	In stock; low stock (1 left); out of stock; zero price; fractional price (e.g. 0.99); high price; 3-decimal price	To confirm
Cart states	Empty; single line; multi-line; quantity at max; very large quantity	To confirm
Coupons	Valid percentage; valid fixed amount; expired; invalid; minimum-spend; 100%; stackable	To confirm
Tax	Taxable; exempt; multiple rates; rate change mid-cart	To confirm
Shipping	Free-shipping threshold; per-weight; per-destination; per-method	To confirm
Currency	Single or multi; 0-dp currency (e.g. JPY) and 2-dp currency	To confirm
Users	Guest; registered; a second registered user for authorization tests	To confirm
Payment	Sandbox success; decline; timeout/cancel; 3-D Secure	To confirm

4.3 Proposed matrices (to be approved)

- **Browsers:** latest Chrome, Edge, Firefox, Safari (and one prior major each).
- **Viewports:** 375 px (mobile), 768 px (tablet), 1440 px (desktop).

5. Risks & Assumptions

Risks

ID	Risk	Impact	Mitigation
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R 1	No acceptance criteria (G1)	Scope disputes; requirements go untested	Obtain ACs or get Section 3 approved as the baseline before execution
R 2	Pricing/rounding defects (G18)	Financial loss and reputational damage	Hand-calculated verification for every money path; confirm server-side authority
R 3	Unnamed integrations (G6)	Shipping and payment scenarios blocked	Identify sandbox endpoints and test modes up front
R 4	"if applicable" conditions (G7–G11)	Untested paths shipped silently	Force an explicit in-scope / out-of-scope decision
R 5	Ticket status "In Review" (G17)	Requirements change mid-testing; rework	Freeze scope at execution start
R 6	No owner on the ticket (G17)	Questions stall; gaps stay unanswered	Assign an owner and a response expectation
R 7	No cart implementation in this repository	Cannot inspect code to confirm behaviour	Request repo/build access, or plan a black-box approach
R 8	Quantity/price trust on the client (G14)	Price tampering, cart manipulation	Confirm server-side recomputation and authorization

Assumptions

- A1: This is a web-based e-commerce cart; behaviour is observed through the browser.
 - A2: The cart is authoritative server-side, and totals are recomputed rather than trusted from the client.
 - A3: Everything listed in the ticket is intended to be tested, even where marked "Low Priority", unless the author defers it (G11).
 - A4: Where the ticket says "if applicable", the default assumption is "in scope, behaviour to be defined" — no scenario is silently dropped.
 - A5: The ticket description is the only requirement source; there is no linked design or spec to cross-check.
 - A6: Money values use a 2-decimal currency unless told otherwise.
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6. Entry / Exit Criteria

Entry

- G1 and G3–G6 answered: acceptance criteria (or Section 3 approved), environment reachable, test data available, integrations usable in sandbox.
- Scope frozen with an explicit decision on the conditional features (G7–G11).
- An owner is assigned who can answer the Section 2 questions (G17).

Exit

- All P0 scenarios pass.

- No open P0 or P1 defects.
 - Subtotal, tax, shipping and total verified against independent hand calculations, with rounding documented.
 - Authorization checks (R-01...R-03) pass, or are explicitly deferred with sign-off.
 - Every gap in Section 2 is either resolved or recorded as an accepted, signed-off risk.
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HUMAN REVIEW GATE

This plan is a draft and is not approved. It stops here for a human to review.

What I assumed (unconfirmed):

- The scenario set is complete relative to the ticket description; nothing was dropped for the "if applicable" items — they are flagged instead.
- Priority ratings (P0/P1/P2) are my proposal from risk, not from the ticket.
- Expected results marked **Spec needed** were deliberately not invented.

Open questions that block sign-off:

- G1 — acceptance criteria, or approval of Section 3 as the baseline.
- G3–G6 — environment, test data, and named shipping/payment sandboxes.
- G13 — quantity bounds, and whether a quantity of 0 is an error or a removal.
- G14 — the authorization model, especially cart-id access control.
- G11 — which advanced features are in this release.
- G17 — who owns the ticket and will answer these questions.

Next step: please confirm, correct or extend the above. In particular, decide the in/out-of-scope boundary for the conditional features and supply the Section 4 environment and test data. I will not write test cases or automation from this plan until a human approves it.

Waiting on approval — treat every priority and expected result above as provisional.